Interpretable and generative models for mouse spaceflight

Jason Trinh
EECS | UC Berkeley

Mentor: James Casaletto | August 2026

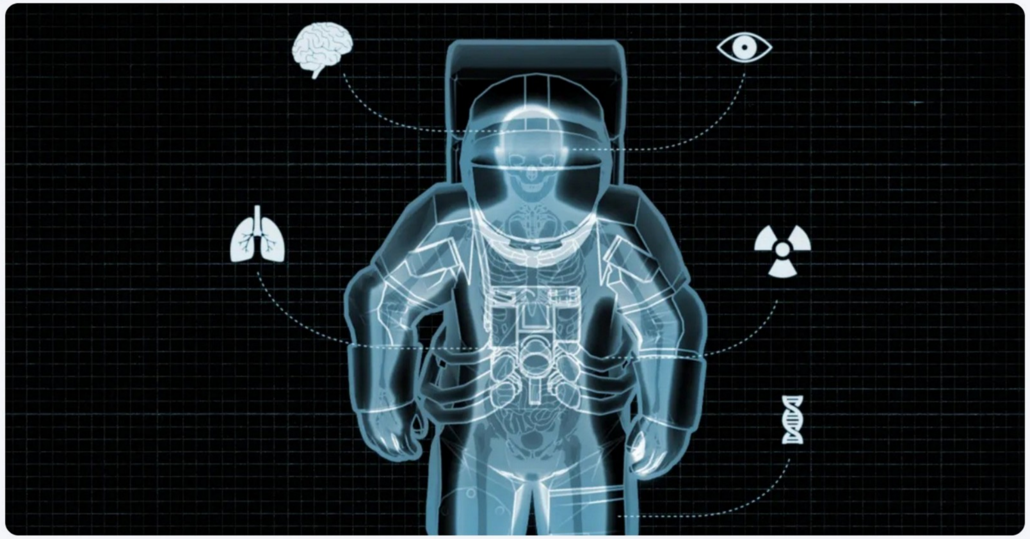
PROJECT OBJECTIVE

Learn how spaceflight changes living systems, then model the biology it produces

Using mouse bulk RNA-seq from NASA missions, we compare spaceflight (FLT) and ground-control (GC) mice across many tissues.

GOAL 1 · UNDERSTAND

Which genes, pathways, and systems does spaceflight alter?



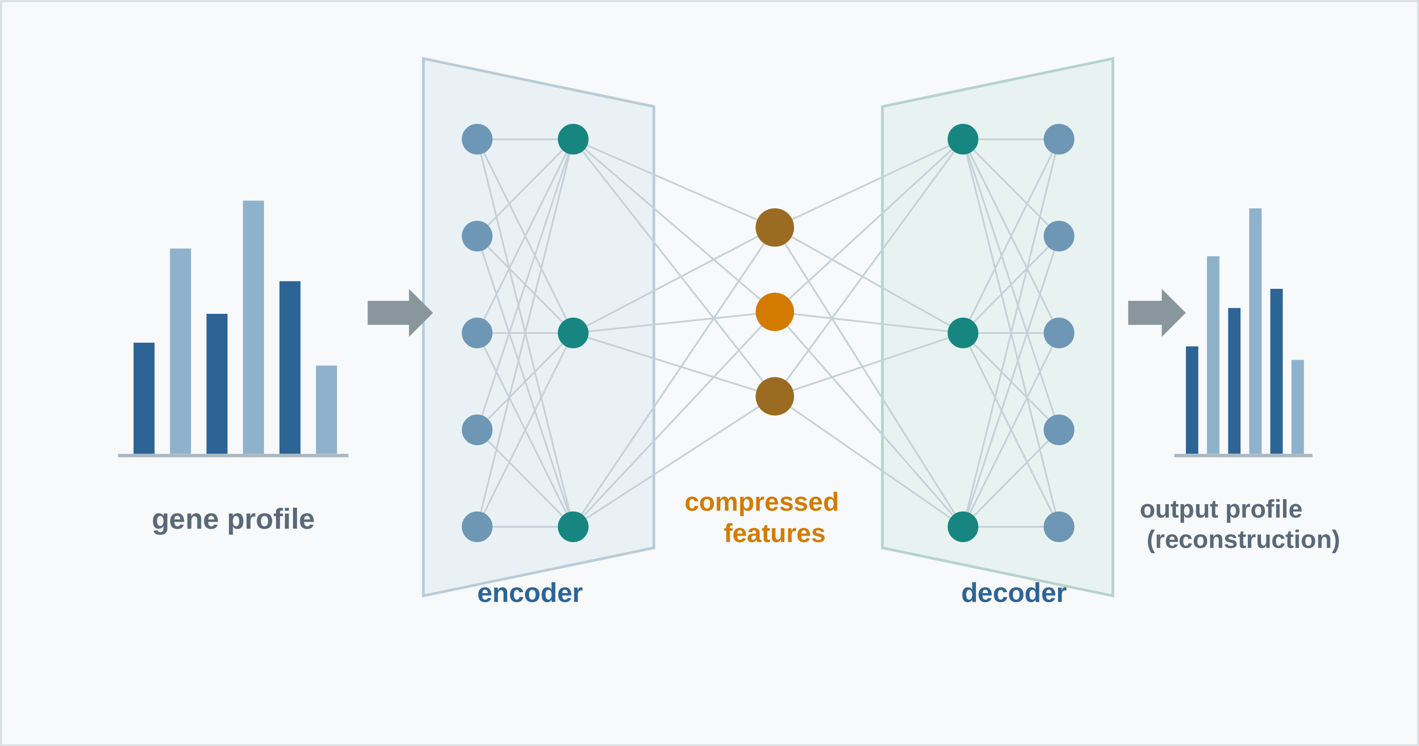
GOAL 2 · MODEL & GENERATE

Learn meaningful representations and generate realistic profiles

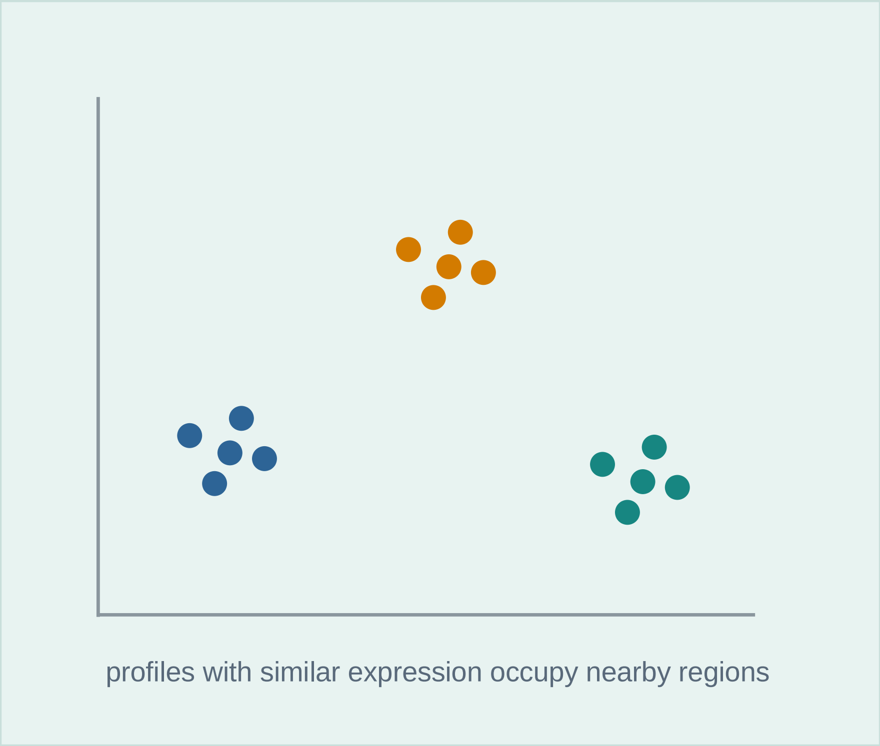
Autoencoders compress thousands of genes into a few features

The encoder summarizes a gene profile; the decoder learns enough structure to reconstruct it.

AUTOENCODER



COMPRESSED SPACE

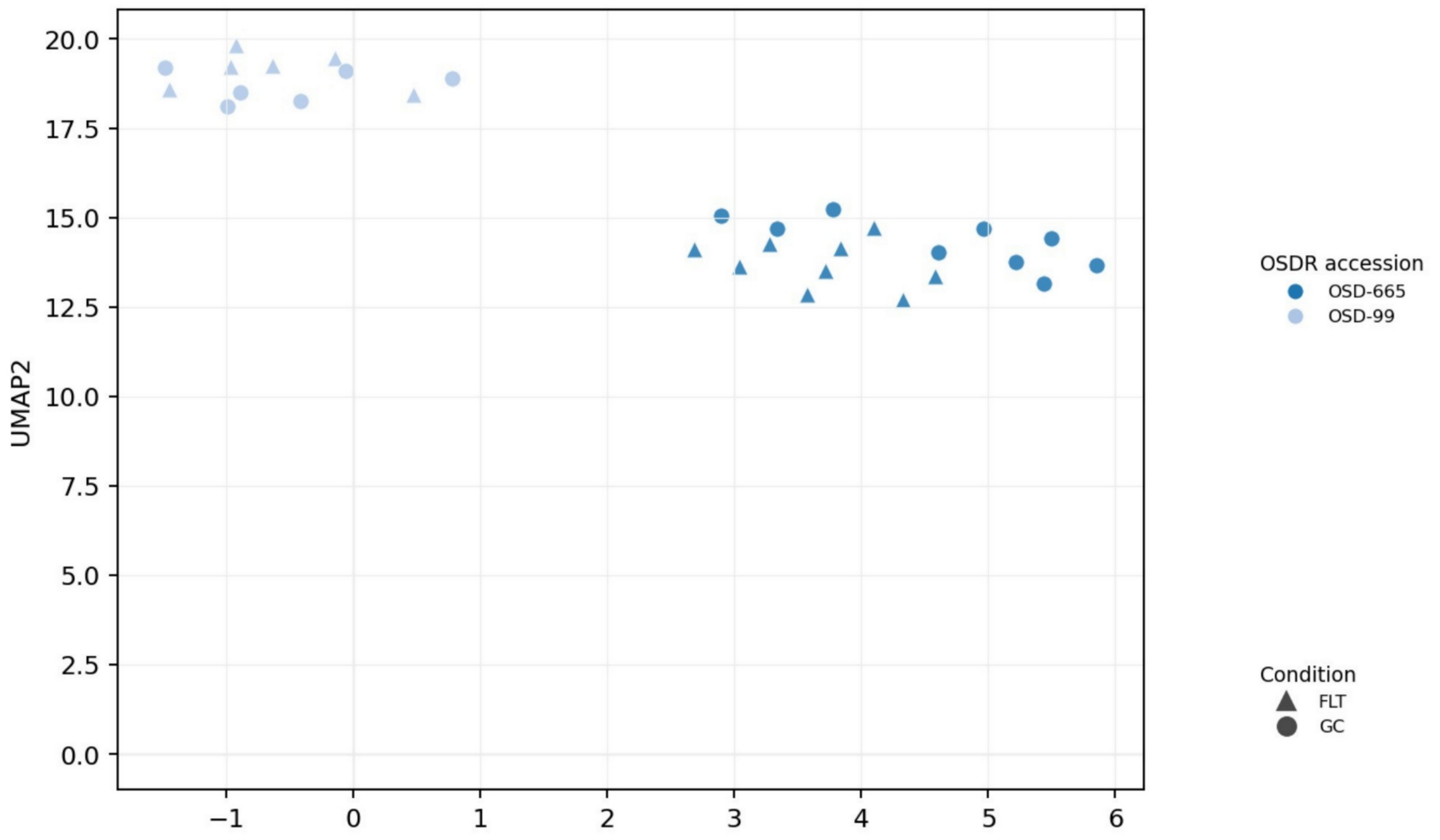


Concept adapted from the midpoint presentation. expiMap: Lotfollahi et al., Nature Cell Biology (2023).

THE COMPLICATION

Study identity dominates the expression structure

In EDL muscle, the two studies separate while FLT and GC overlap within each study.



Midpoint slide 7; EDL profiles from OSD-665 and OSD-99.

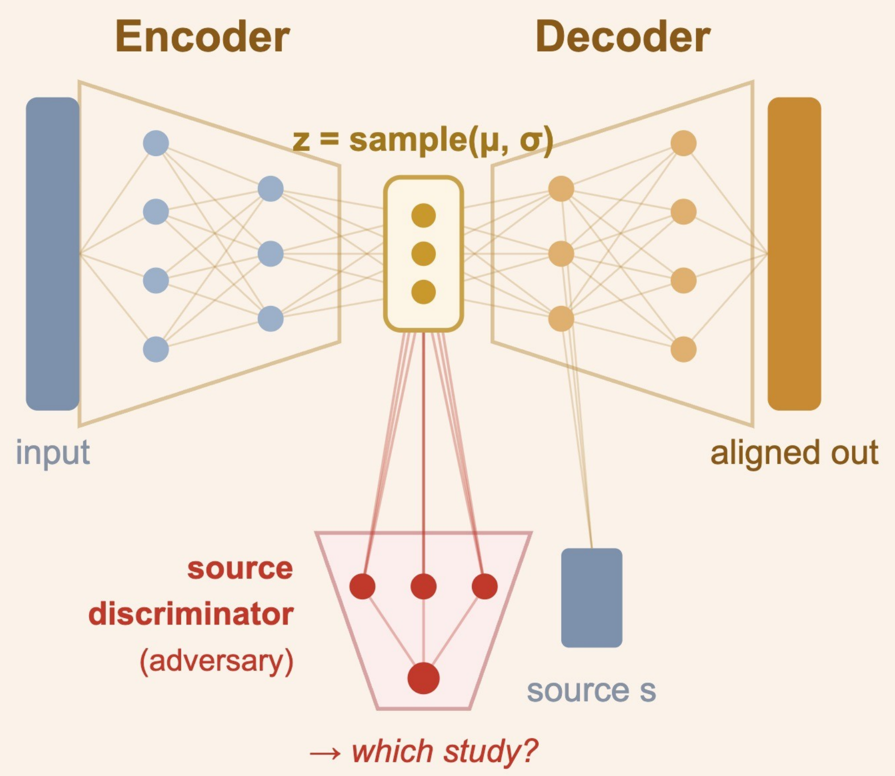
THE MAIN CHALLENGE

Can we just subtract the study effect?

MOBER (Multi-Origin Batch Effect Remover) is a deep domain-adaptation model that pulls studies into one distribution while trying to preserve biology.

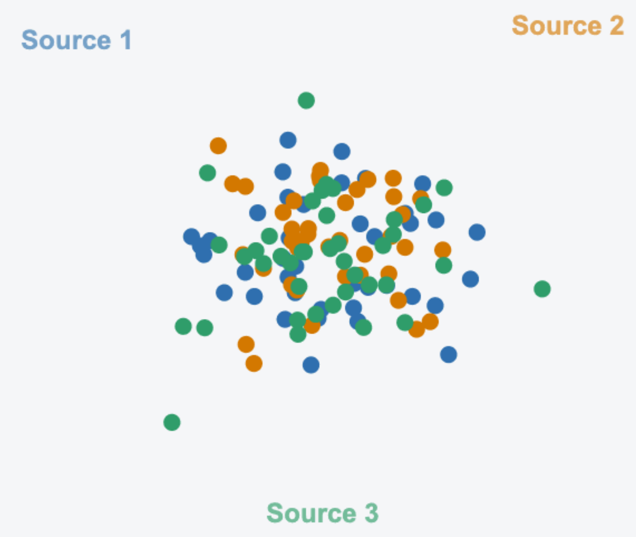
How MOBER works

VAE encoder–decoder + source-discriminator adversary



What it does · align studies

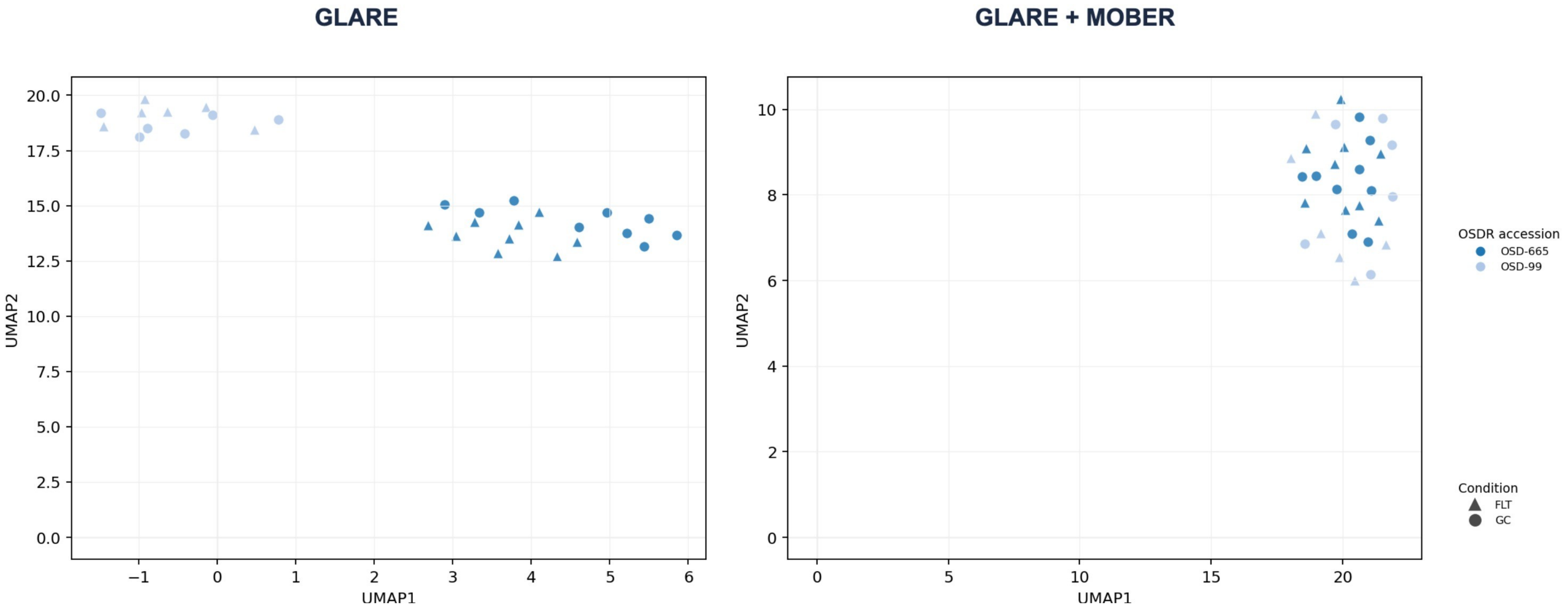
pull separate studies into one shared distribution



THE MAIN CHALLENGE · EVIDENCE

MOBER merges the two muscle studies, but flight and ground still overlap

Skeletal muscle (EDL): here MOBER **does** pull the two studies together, but FLT vs GC still doesn't separate, and the correction is only partial.



SILHOUETTE SCORES

Study separation

0.531 → 0.179 ▼ better

GLARE → + MOBER

Flight separation

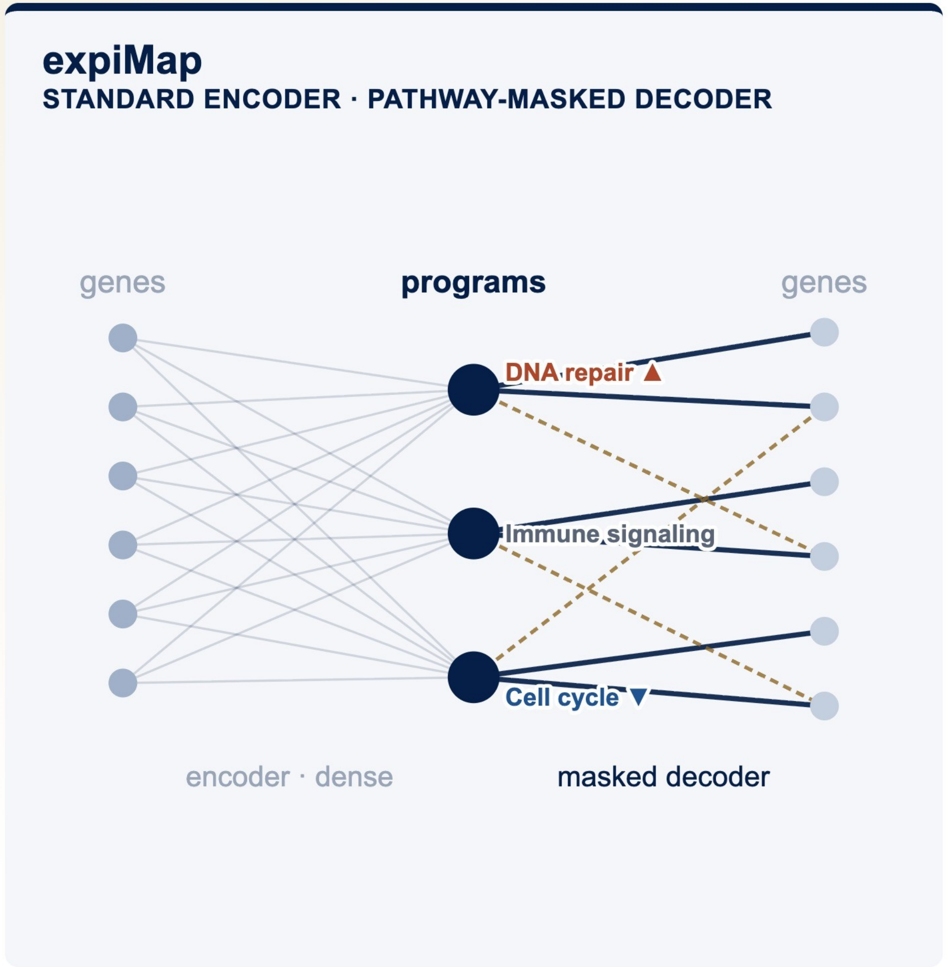
0.002 → −0.002 ≈ 0

GLARE → + MOBER

METHOD SPOTLIGHT

Don't learn the biology, wire it in

GLARE and MOBER tried to **learn** structure from the data. **expiMap** flips it: known Reactome pathways are **built into** the decoder, so each latent axis **is** biology by construction.



WHY IT'S INTERPRETABLE

- Each latent node is one Reactome program (~1,140).
- A soft mask lets a stray gene help only when the data support it.
- Compare FLT and GC one program at a time.

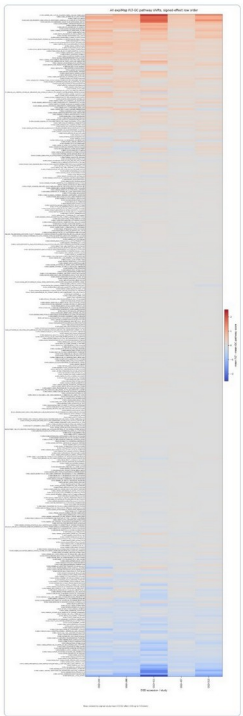
EXPIMAP SCORES

Program scores summarize pathway changes within one tissue

	FLT score	-	GC score	=	FLT - GC
Extracellular matrix	+0.47	-	+0.08	=	+0.39
Cell-cycle control	+0.42	-	+0.10	=	+0.32
Immune signaling	+0.15	-	+0.13	=	+0.02
Oxidative metabolism	-0.06	-	-0.03	=	-0.03
DNA repair	-0.10	-	+0.34	=	-0.44

lower · little change · higher

OUTPUT · PROGRAMS × SAMPLES



thymus · red ↑ / blue ↓ in FLT

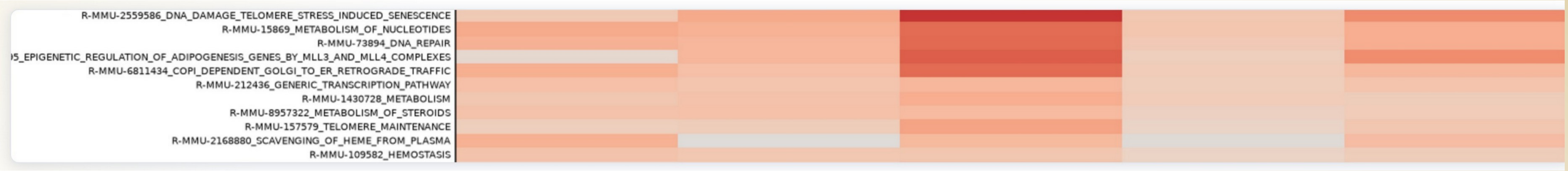
Example values, ordered from higher to lower FLT - GC. Literature review appears next.

EXPIMAP INTERPRETATION

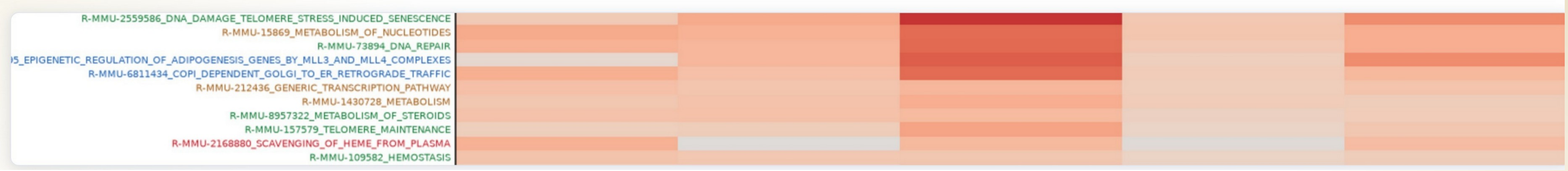
Each program, reviewed against prior literature

The pathway scores stay fixed; the pathway-name colors summarize the literature comparison.

1 PATHWAY SCORES



2 LITERATURE LABELS



Aligned **Complementary** **Uncertain** **Conflicting** **Low effect**

Name color records the literature assessment; heatmap color records the program score.

Heatmaps reproduced from midpoint slides 11 and 12.

EXPIMAP RESULTS

Five tissues showed recurring pathway patterns

FLT and GC were compared within each study before the study-level changes were combined.

Tissue	Direction / pathways	Interpretation
Thymus 117 samples | 5 projects	LOWER IN FLIGHT **DNA repair; RHOA cytoskeletal cycle; lymphoid-stromal interactions**	Known involution may also involve lower repair, motility, and niche coordination.
Skin 151 samples | 4 projects	LOWER IN FLIGHT **Chromatin regulation; DNA repair; Hedgehog; sphingolipids; cell junctions**	Barrier injury may include a broader loss of tissue maintenance and coordination.
Liver 197 samples | 9 projects	LOWER IN FLIGHT **MHC class II antigen presentation; T-cell receptor signaling**	Metabolic heterogeneity coexisted with lower adaptive immune communication.
Spleen 100 samples | 5 projects	LOWER IN FLIGHT **T-cell receptor signaling; neutrophil degranulation; C-type lectin signaling**	The most consistent multi-pathway result joined adaptive and innate immune changes.
Kidney 135 samples | 6 projects	HIGHER IN FLIGHT **ECM proteoglycans; WNT signaling; IGF transport and uptake**	Matrix and growth-factor programs rose together, suggesting a renal remodeling response.

Source: expiMap manuscript Figures 3, 4, and 6. Program-score direction does not by itself prove biochemical activation or inhibition.

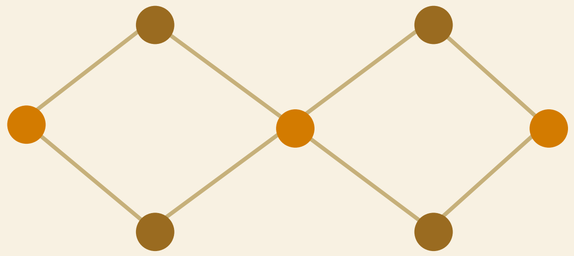
What is synthetic transcriptomics?

A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

01 Observed profiles

Measured gene expression from real mouse tissue

02 Learn expression patterns



Capture gene relationships and variation by tissue and FLT/GC condition

03 Synthetic profiles

Sample new numeric expression vectors for a selected context

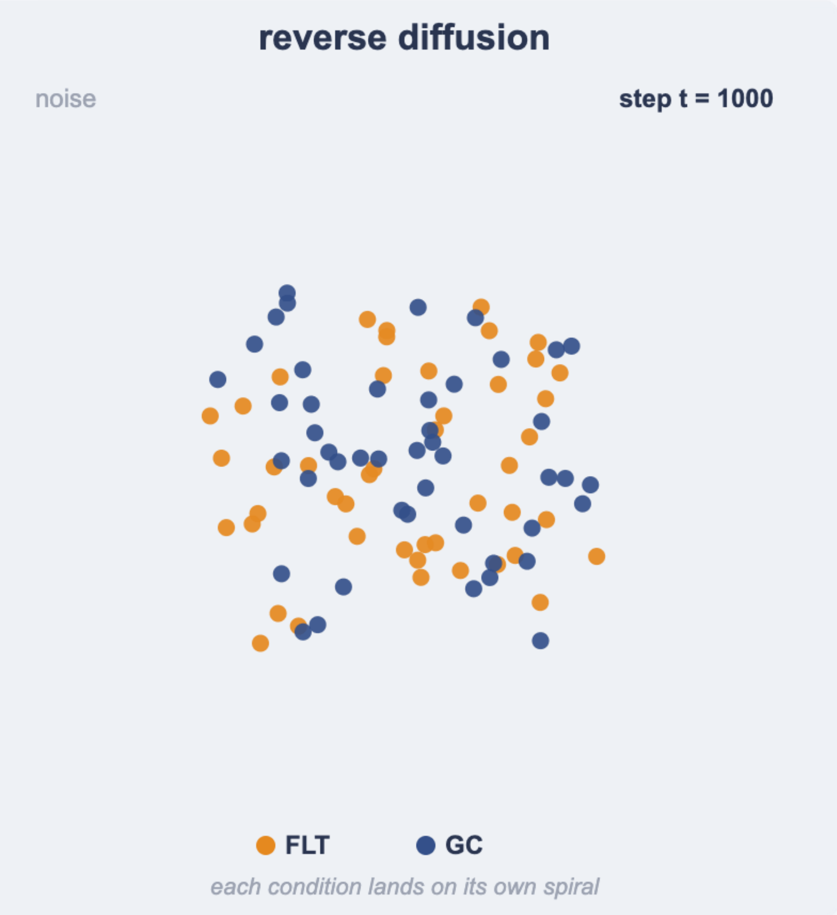
HOW THE GENERATORS WORK · 1 OF 2

Diffusion: denoise from noise

Start from pure noise and denoise step by step onto the data manifold, conditioned on FLT or GC.

FAMILIAR · IMAGE DIFFUSION

OURS · GENE-EXPRESSION DIFFUSION

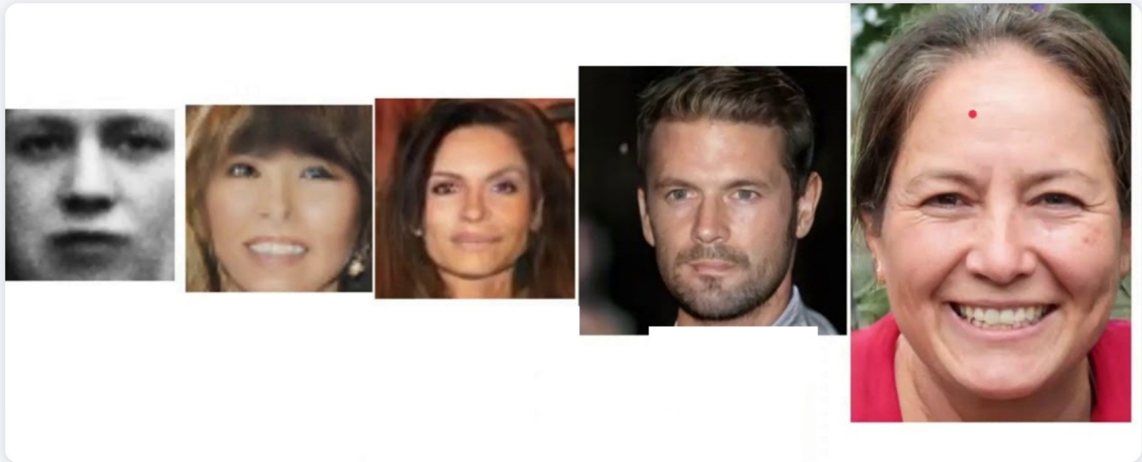


HOW THE GENERATORS WORK · 2 OF 2

Conditional WGAN-GP: generator versus critic

A generator and a Wasserstein critic compete until fake profiles are indistinguishable from real.

FAMILIAR · GANS FOR FACES



OURS · PROFILE GENERATOR

MODEL DEVELOPMENT

Building the RNA-seq generator

We compared how to process the data, handle study differences, train the model, and tell it what to generate.

01 Data scope

Sources

- **OSDR API**
- **ARCHS4**

Studies

- Single
- **Multiple**

Tissues

- Per tissue
- **All tissues**

02 Processing

Expression

- Raw
- CPM
- **TPM**

Scaling

- None
- Z-score
- **MaxAbs**

Features

- All genes
- HVGs
- Reactome
- **L1000 map**

03 Study effects

Alternative methods

- **None**
- Within-study z-score
- ComBat / MBatch
- MOBER

04 Model training

Generator

- WGAN-GP
- **DDIM**

Training source

- OSDR only
- ARCHS4 only
- **ARCHS4 then OSDR**

05 Conditioning

Model inputs

- **Tissue**
- **FLT / GC**
- **Study**
- **Material type**
- Sex / age

Downstream configuration

PROCESS: **TPM / MaxAbs / 974 landmarks** → PRETRAIN: **ARCHS4** → ADAPT: **OSDR** → GENERATE: **Conditional DDIM**

CONDITION: **Tissue** | **FLT / GC** | **Study** | **Material type**

HARMONIZE: **None**

SCOPE: **All tissues**

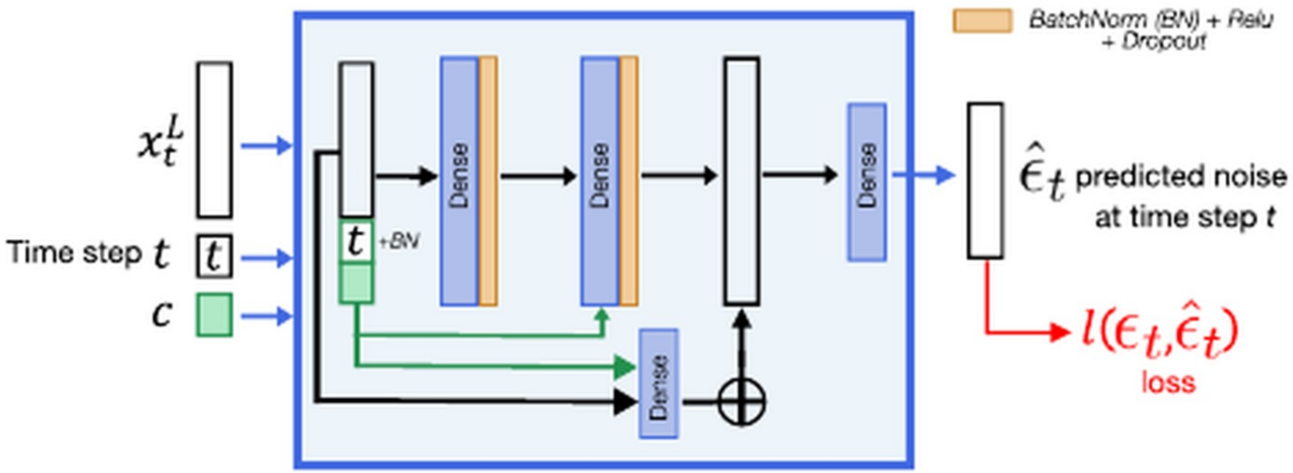
Pipeline evaluated with OSDR and ARCHS4. Generator designs follow Vinas et al. (2022) and Lacan et al. (2026).

VALIDATION

Diffusion best reproduced the measured expression distribution

Both models reproduced gene expression. Diffusion samples were harder to distinguish from measured samples.

Architecture from Lacan et al.



The network repeatedly predicts and removes noise using tissue and time information.

For OSDR, the model also receives FLT/GC, study, and sample-material information.

Tissue identity retained Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

Metric	WGAN-GP	DDIM	Read
Expression corr.	0.976	0.974	higher
Coverage (F1)	0.985	0.997	higher
Real-vs-synth. acc.	0.636	0.475	0.5 is ideal
Distribution distance	0.144	0.074	lower

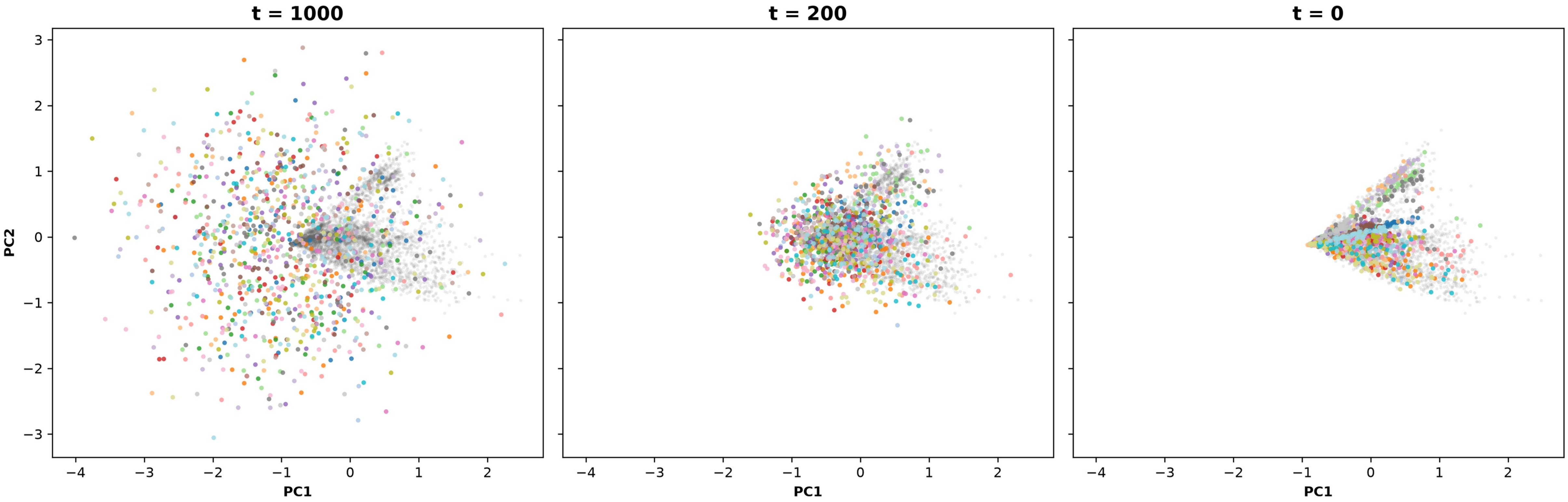
Use diffusion Better coverage and closer to the real distribution.

A score near 0.5 means a prediction model cannot reliably tell measured and generated samples apart.

Architecture: Lacan et al. (2026), Fig. 1C, doi:10.1186/s12859-026-06470-8. Each model was evaluated on its own fixed test split.

Diffusion learns tissue structure from noise

The same PCA axes follow 1,024 generated profiles through reverse diffusion.



t = 1000: noise

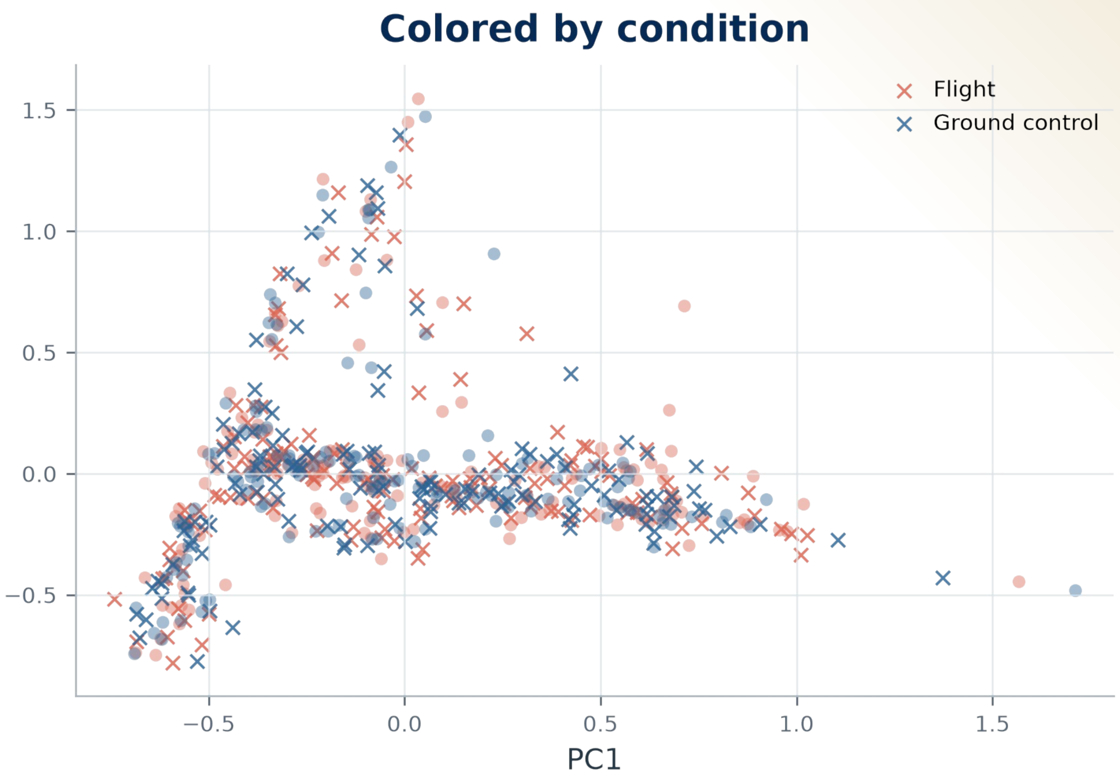
t = 200: partial structure

t = 0: tissue-conditioned profiles

Gray points are real ARCHS4 profiles; colors identify generated tissue conditions. PC1 and PC2 limits are shared across panels.

FLT and GC overlap in the global PCA view

The condition signal is not clear when all tissues and studies are combined.

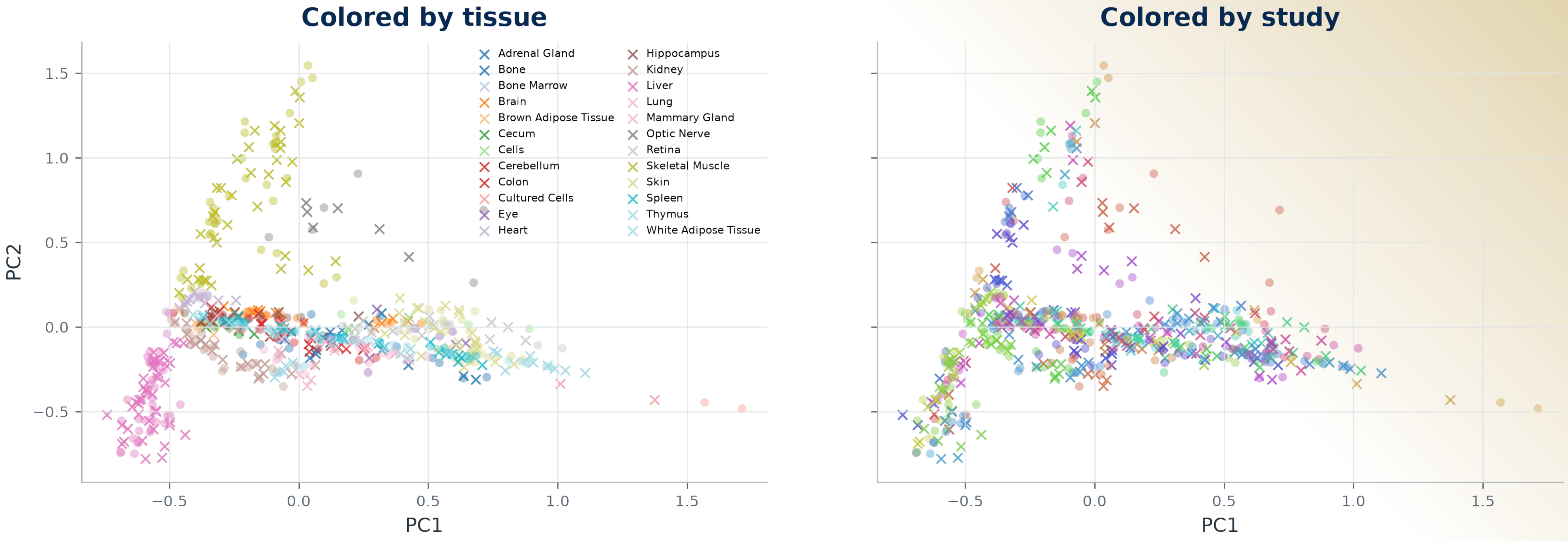


Color: FLT or GC. Circles are real samples; crosses are generated samples.

DIFFUSION OUTPUT

Tissue and study structure dominate the PCA space

We therefore compare FLT and GC within tissues and account for study.



Left: tissue. Right: study. Circles are real samples; crosses are generated samples.

PRIMARY ANALYSIS

Does adding synthetic data improve FLT vs GC prediction?

Train the same tissue-specific classifier two ways, then compare both versions on real samples.

01 **TRAIN ON REAL DATA**

R Real profiles → FLT / GC classifier →

02 **TRAIN ON REAL + SYNTHETIC DATA**

R + S Real + synthetic profiles → FLT / GC classifier →

TEST BOTH

Same real samples

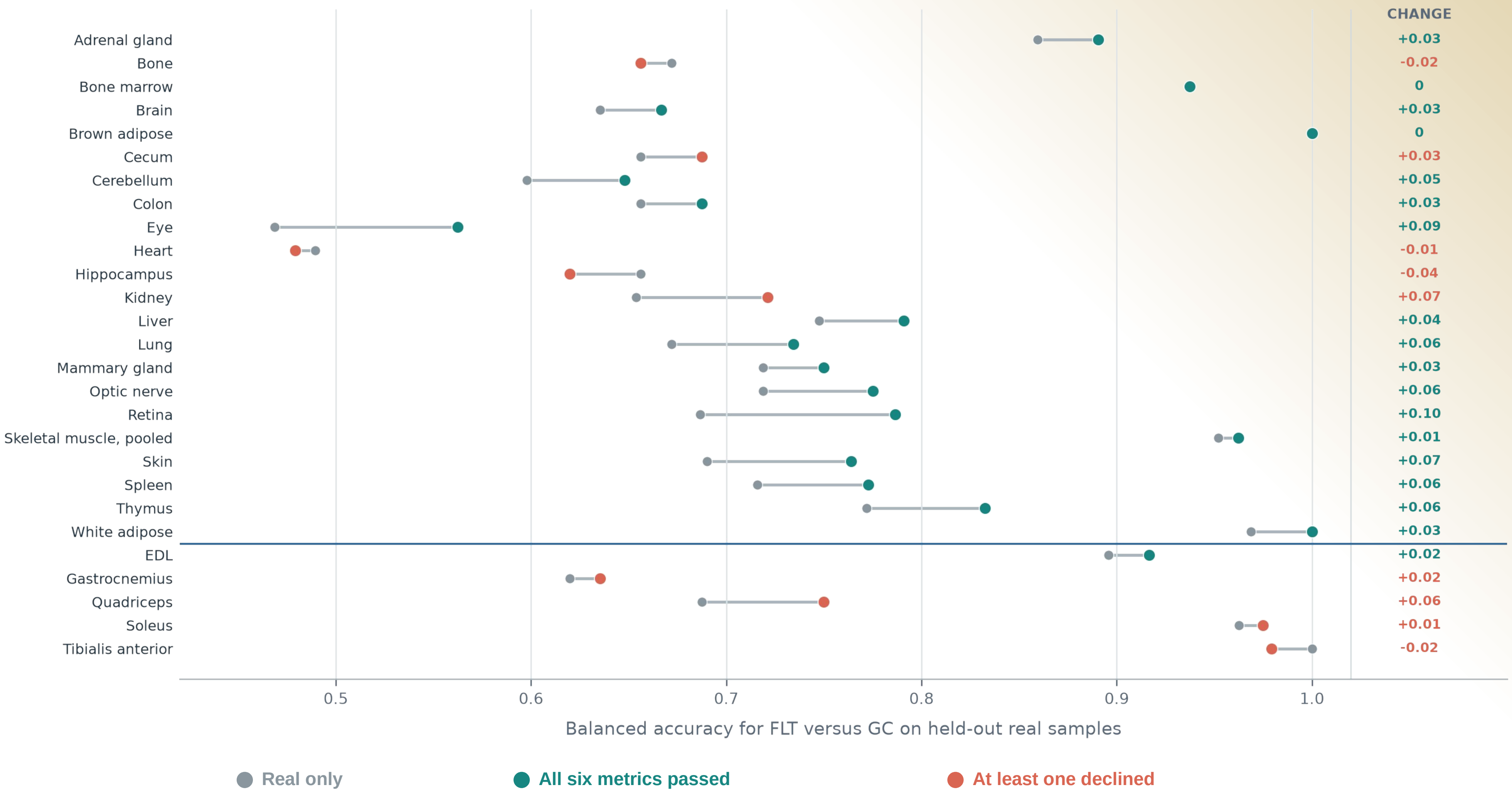
FLT GC

Compare prediction accuracy

Both classifiers are evaluated on real OSDR profiles.

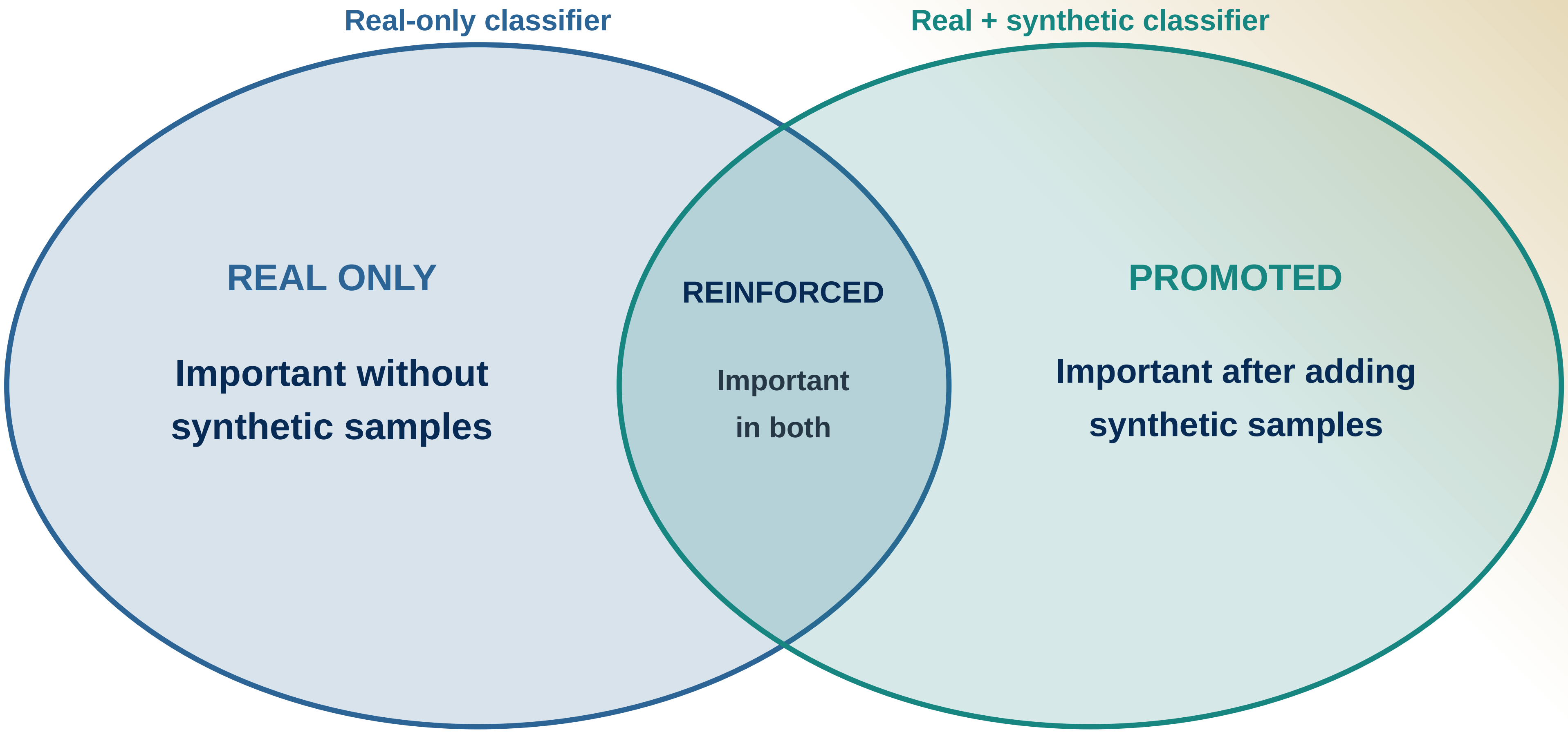
PREDICTION

Real + synthetic vs real-only balanced accuracy across 27 tissues



Outcome: FLT versus ground control within each tissue. All versions used the same 974 genes and prediction settings.

Compare what each classifier finds important



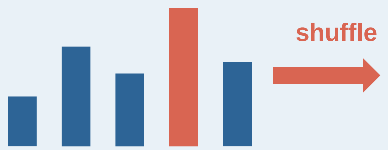
SYNTHETIC ANALYSIS

We compared feature importance at three levels

Each analysis compares the real-only and real + synthetic classifiers.

1 INDIVIDUAL GENES

Permutation + SHAP



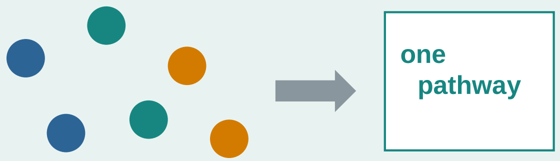
Permutation: does prediction weaken?

SHAP: does the gene push toward FLT or GC?

OUTPUT · RANKED GENES

2 REACTOME GROUPS

Grouped permutation + grouped SHAP



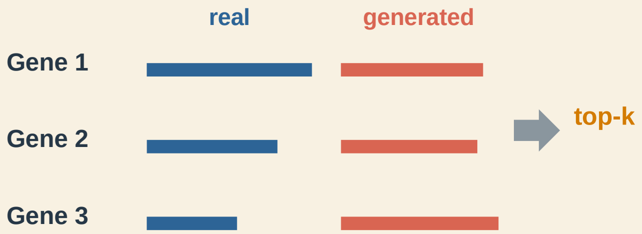
Move related genes together

Test correlated genes as one group

OUTPUT · RANKED PATHWAYS

3 CONSENSUS RANKING

Compact gene panels



Combine measured and generated ranks

Test small panels of the highest-ranked genes

OUTPUT · COMPACT GENE PANELS

RESULTS

The three analyses retained different types of results

Every retained candidate was also supported by the real OSDR data.

Individual genes

Permutation + SHAP

21 associations in 4 tissues

7 promoted | 14 reinforced

RETAINED BY TISSUE

Thymus	15
Liver	4
Skin	1
Spleen	1

Reactome groups

Grouped permutation + SHAP

10 pathways in 3 tissues

4 promoted | 6 reinforced

RETAINED BY TISSUE

Thymus	7
Skin	2
Spleen	1

Consensus ranking

Compact gene panels

49 associations in 10 analyses

26 promoted | 23 reinforced

RETAINED BY TISSUE

Thymus	16
Pooled muscle	12
Soleus	5
Seven others	16

Promoted: retained only after adding synthetic information. Reinforced: retained in both analyses.

RESULTS

All three analyses found candidates in thymus, skin and spleen

TISSUE	Individual genes Permutation + SHAP	Reactome groups Grouped permutation + SHAP	Consensus ranking Compact gene panels
Thymus	15 | 7 promoted, 8 reinforced	7 | 2 promoted, 5 reinforced	16 | 13 promoted, 3 reinforced
Skin	1 | 0 promoted, 1 reinforced	2 | 2 promoted, 0 reinforced	1 | 1 promoted, 0 reinforced
Spleen	1 | 0 promoted, 1 reinforced	1 | 0 promoted, 1 reinforced	4 | 3 promoted, 1 reinforced
Liver	4 | 0 promoted, 4 reinforced	-	-
Pooled muscle	-	-	12 | 4 promoted, 8 reinforced
Soleus	-	-	5 | 0 promoted, 5 reinforced
Kidney	-	-	2 | 1 promoted, 1 reinforced
Tibialis anterior	-	-	4 | 1 promoted, 3 reinforced
Gastrocnemius	-	-	2 | 2 promoted, 0 reinforced
Adrenal gland	-	-	2 | 1 promoted, 1 reinforced
Eye	-	-	1 | 0 promoted, 1 reinforced

FOUND BY ALL THREE

Results below the divider came from one analysis.

Counts refer to tissue-feature associations. Genes and pathways are different evidence units.

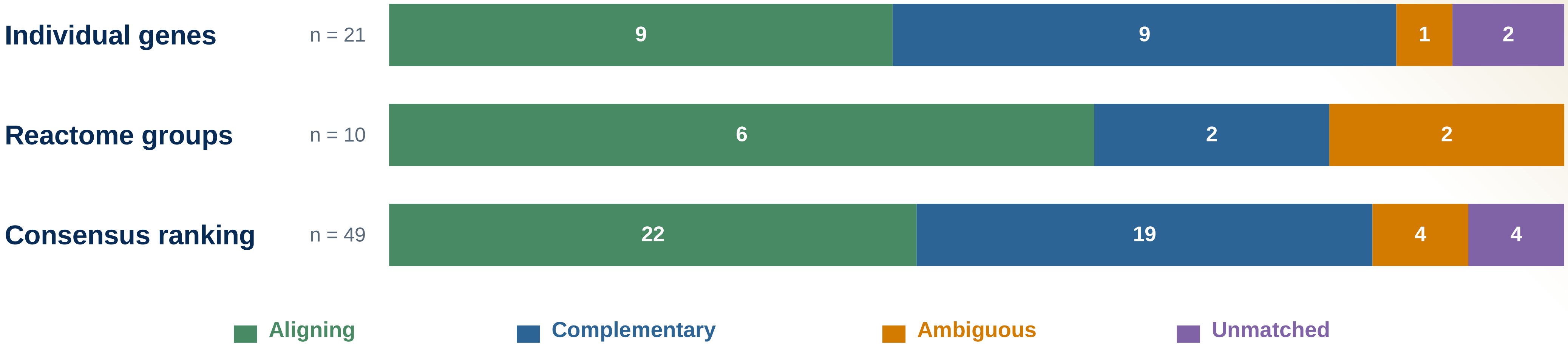
LITERATURE REVIEW

Feature selection and literature review answer separate questions

Selection compares classifiers. Literature review compares each candidate with prior evidence.

MODEL COMPARISON	Real-only importance →	Real + synthetic importance →	**Promoted or reinforced**
EVIDENCE REVIEW	Candidate + tissue + direction →	Search prior studies →	**Evidence category**

Literature classifications among retained candidates

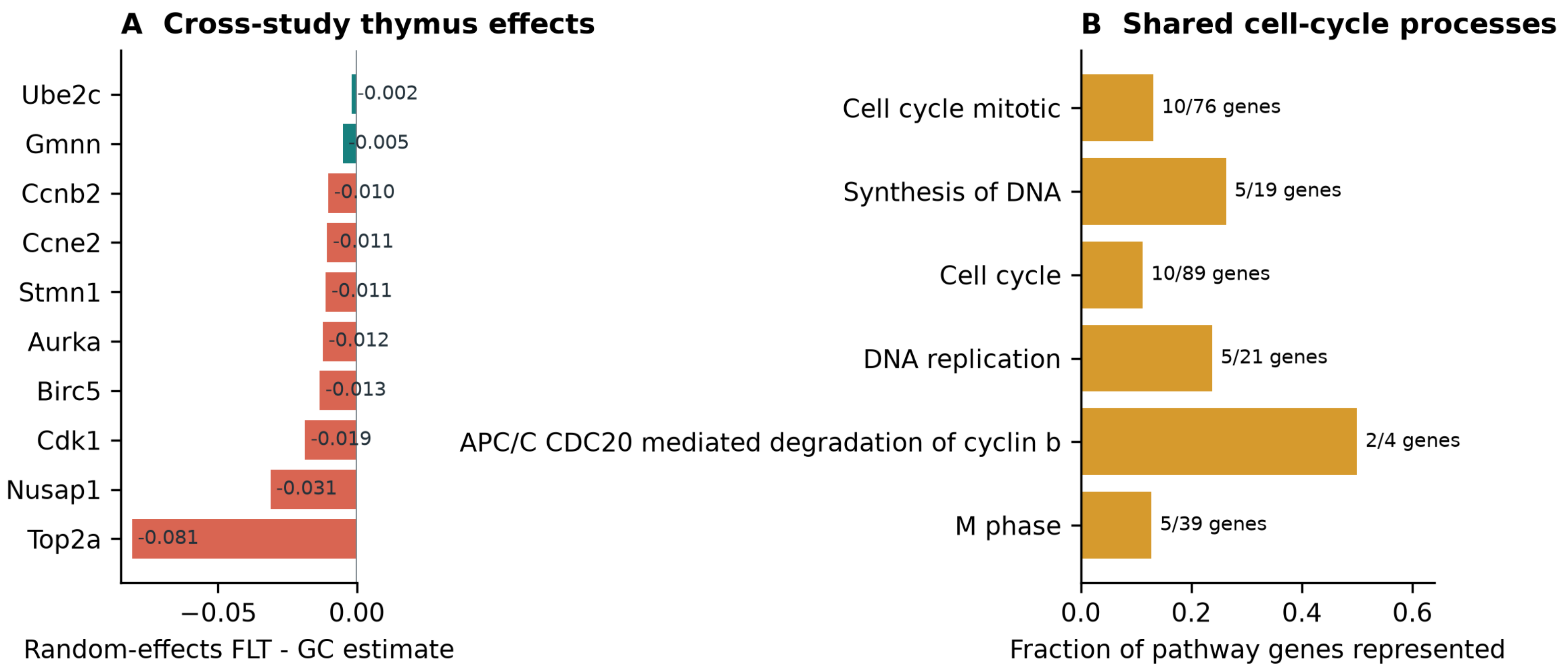


Annotations used the fixed tissue, FLT-GC direction and candidate identity; detailed rationale and sources are in Tables S16, S22 and S23.

Thymus: all three analyses point to lower cell-cycle activity in flight

Genes, Reactome pathways and the compact panel all show the same direction.

Real OSDR effects and shared Reactome processes



INDIVIDUAL GENES

15 **7 promoted | 8 reinforced**

Most retained genes are FLT lower.

REACTOME GROUPS

7 **2 promoted | 5 reinforced**

Six pathways describe mitosis, DNA replication or APC/C control.

CONSENSUS RANKING

16 **13 promoted | 3 reinforced**

Thirteen genes are FLT lower, including CDK1, TOP2A and AURKA.

BIOLOGICAL READING

Lower thymic proliferative activity

Individual genes: Table S19. Reactome groups: Table S23. Consensus ranking: Table S10.

BIOLOGICAL INTERPRETATION

Skin and spleen show different flight-associated patterns

Skin: regulated cell death | Spleen: structural remodeling

SKIN

Interferon-linked regulated cell death

GENE EVIDENCE	PLSCR1 was higher in flight and retained by two analyses
PATHWAY EVIDENCE	Two necroptosis pathways were higher in flight
CONTEXT	PLSCR1 is interferon inducible

COMPLEMENTARY HYPOTHESIS

Direct spaceflight evidence for skin necroptosis is limited

SPLEEN

Matrix, adhesion and cytoskeletal remodeling

GENE EVIDENCE	LOXL1 was retained by two analyses; RAI14, PTPRK and MYL9 were also flight higher
PATHWAY EVIDENCE	No shared structural Reactome pathway
CONTEXT	The genes span matrix, adhesion and cytoskeletal functions

COMPLEMENTARY HYPOTHESIS

Matrix, adhesion and cytoskeletal remodeling

All displayed gene results were higher in flight. Direct spaceflight evidence for both complementary hypotheses was limited.

TAKEAWAYS

What the models can and cannot tell us

01 Computational evidence

expiMap and the generative models summarize patterns in the measured RNA-seq data. They do not reveal mechanism directly.

02 Candidates to follow up

Agreement among genes, pathways, models, and prior studies narrows the list of tissues and targets.

03 Experimental testing

Wet-lab experiments must confirm the biological changes and distinguish regulation from shifts in cell composition.

Thank you

Questions?

Jason Trinh

jasontrinh@berkeley.edu

Mentor	**James Casaletto**
Program	NASA Space Life Sciences Training Program
Data	NASA OSDR | ARCHS4 | Reactome
AI tools	ChatGPT | Claude
Code and manuscript	github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.